Example

Take $W = \text{span}\left\{\begin{bmatrix}1\\i\end{bmatrix}\right\} \subseteq \mathbb{C}^2$.

Compute $W^\perp = \{ v \in \mathbb{C}^2 : \langle v, w\rangle = 0 \ \forall w \in W\}$.

Exercise: $v \in W^\perp \iff v$ orthogonal to spanning set for W.

Therefore $v = \begin{bmatrix}x\\y\end{bmatrix} \in \mathbb{C}^2$ belongs to $W^\perp$

$$\iff \left\langle \begin{bmatrix}x\\y\end{bmatrix}, \begin{bmatrix}1\\i\end{bmatrix}\right\rangle = 0$$

$$\iff (x)\overline{(1)} + (y)\overline{(i)} = 0 \qquad \overline{i} = \overline{(0+1i)} = 0 - 1i = -i$$

$$\iff x + (y)(-i) = 0$$

$$\iff x - yi = 0$$

$$\iff x = yi$$

$$\iff v = \begin{bmatrix}yi\\y\end{bmatrix} = y\begin{bmatrix}i\\1\end{bmatrix}$$

$$\iff v \in \text{span}\left\{\begin{bmatrix}i\\1\end{bmatrix}\right\}.$$

Therefore $W^\perp = \text{span}\left\{\begin{bmatrix}i\\1\end{bmatrix}\right\}$

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§6.3 Orthogonal sets

Def. Let $\{u_1, \dots, u_k\}$ be a finite set of vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$. The set $\{u_1, \dots, u_k\}$ is <u>orthogonal</u> if $\langle u_j, u_\ell\rangle = 0$ for all $1 \le j \ne \ell \le k$.

Example

Recall $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $\ldots$, $e_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$.

(the standard basis vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$).

It is easy to see that

$$\langle e_1, e_2 \rangle = \cancel{(1)(0)} + \cancel{0(1)} + (0) + \cdots + 0$$
$$= 0.$$

Similarly, for any $j \neq l$, $\langle e_j, e_l \rangle = 0 + \cdots + 0 = 0$.

It follows that $\{e_1, \ldots, e_n\}$ is an orthogonal set.

Similarly $\{e_1, e_2, e_3\}$ is also an orthogonal set.

Any nonempty subset of $\{e_1, \ldots, e_n\}$ is orthogonal.

Exercise: Any nonempty subset of an orthogonal set is still orthogonal.

Clearly the zero vector is orthogonal to any vector

i.e. $\langle v, 0 \rangle = \langle 0, v \rangle = 0$.

$$\langle v, 0 \rangle = \left\langle v, \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \right\rangle$$

Exercise: If $\{u_1, \ldots, u_k\}$ is orthogonal then so is $\{u_1, \ldots, u_k, 0\}$.

Example 1 in textbook.

Take $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}$.

Let's show that $\{u_1, u_2, u_3\}$ is an orthogonal set in $\mathbb{R}^3$.

$$\langle u_2, u_1 \rangle = \langle u_1, u_2 \rangle = (3)(-1) + (1)(2) + (1)(1)$$
$$= -3 + 2 + 1 = 0.$$

$$\langle u_3, u_2 \rangle = \langle u_2, u_3 \rangle = (-1)(-1) + (2)(-4) + (1)(7)$$
$$= 1 - 8 + 7 = 0$$

$$\langle u_3, u_1 \rangle = \langle u_1, u_3 \rangle = (3)(-1) + (1)(-4) + (1)(7)$$
$$= -3 - 4 + 7 = 0.$$

Done.

$\{u_1, u_2\}$, $\{u_2, u_3\}$, $\{u_1, u_3\}$ are all orthogonal.

Theorem 4

Suppose $S = \{u_1, \dots, u_k\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$. Then S is linearly ind. and S is a basis for $\operatorname{Span}\{u_1, \dots, u_k\}$.

Sketch of proof.

Suppose $c_1 u_1 + \dots + c_k u_k = 0$ for some $c_1, \dots, c_k$ in $\mathbb{R}$ (or $\mathbb{C}$).

Taking the inner product with u_1,

$$\langle c_1 u_1 + \cdots + c_k u_k, u_1 \rangle = \langle 0, u_1 \rangle$$

$$c_1 \langle u_1, u_1 \rangle + c_2 \cancel{\langle u_2, u_1 \rangle}^{0} + \cdots + c_k \cancel{\langle u_k, u_1 \rangle}^{0} = 0$$ by orthogonality.

$$\Rightarrow \quad c_1 \langle u_1, u_1 \rangle = 0$$

$$\Rightarrow \quad c_1 \|u_1\|^2 = 0$$ Since $u_1 \neq 0$, $\|u_1\| \neq 0$

$$\Rightarrow \quad c_1 = 0.$$

Exercise: Taking the inner product with u_j instead of u_1 gives us $c_j = 0$ for any $1 \leq j \leq k$.

i.e. $c_1 = c_2 = \cdots = c_k = 0$.

and S is lin. ind.

Example Taking $u_1, u_2, u_3 \in \mathbb{R}^3$ as in the previous example, we see from Thm 4 that $\{u_1, u_2, u_3\}$ is lin. ind.

By Thm 13 §4.5, $\{u_1, u_2, u_3\}$ is an orthogonal basis for $\mathbb{R}^3$, since $\dim \mathbb{R}^3 = 3$.

Theorem 5

Suppose $B = \{v_1, \ldots, v_k\}$ is an orthogonal basis for $W = \operatorname{span}\{v_1, \ldots, v_k\}$ in $\mathbb{R}^n$ or $\mathbb{C}^n$.

If $y \in W$ then

$$y = \frac{\langle y, v_1\rangle}{\langle v_1, v_1\rangle} v_1 + \frac{\langle y, v_2\rangle}{\langle v_2, v_2\rangle} v_2 + \cdots + \frac{\langle y, v_k\rangle}{\langle v_k, v_k\rangle} v_k$$

and $$p_B(y) = \begin{bmatrix} \frac{\langle y, v_1\rangle}{\langle v_1, v_1\rangle} \\ \vdots \\ \frac{\langle y, v_k\rangle}{\langle v_k, v_k\rangle} \end{bmatrix}$$ in $\mathbb{R}^k$ or $\mathbb{C}^k$.

Sketch of proof.

Suppose $y = c_1 v_1 + \cdots + c_k v_k$, for some $c_1, \ldots, c_k$ scalars.

Taking the inner product with v_1, we obtain

$$\langle y, v_1\rangle = \langle c_1 v_1 + \cdots + c_k v_k, v_1\rangle$$

$$\Rightarrow \langle y, v_1\rangle = c_1 \langle v_1, v_1\rangle + c_2 \cancelto{0}{\langle v_2, v_1\rangle} + \cdots + c_k \cancelto{0}{\langle v_k, v_1\rangle}$$

$$\Rightarrow \langle y, v_1\rangle = c_1 \langle v_1, v_1\rangle$$

$$\Rightarrow c_1 = \frac{\langle y, v_1\rangle}{\langle v_1, v_1\rangle}$$

Exercise: Taking inner product with v_j gives us $c_j = \dfrac{\langle y, v_j\rangle}{\langle v_j, v_j\rangle}$

as desired.

Example

Take $B = \{u_1, u_2, u_3\}$ as in previous examples. B is an orthogonal basis.

Take $y = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Then by Thm 5

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \frac{\left\langle \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$+ \frac{\left\langle \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} \right\rangle} \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}$$

$$= \frac{2}{11} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} + \frac{-2}{6} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} + \frac{-8}{66} \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix}$$

$$\Rightarrow p_B\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2/11 \\ -2/6 \\ -8/66 \end{bmatrix} \in \mathbb{R}^3.$$

Example

Take $v_1 = \begin{bmatrix} i \\ i \end{bmatrix}$, $v_2 = \begin{bmatrix} 2i \\ -2i \end{bmatrix}$

Then $\langle v_1, v_2 \rangle = \left\langle \begin{bmatrix} i \\ i \end{bmatrix}, \begin{bmatrix} 2i \\ -2i \end{bmatrix} \right\rangle$

$$= (i)\overline{(2i)} + (i)\overline{(-2i)}$$

$$= (i)(-2i) + (i)(2i)$$

$$= 2 + (-2)$$
$$= 0 .$$

Moreover $\langle v_2, v_1 \rangle = \overline{\langle v_1, v_2 \rangle} = \overline{0} = 0$.

Therefore $\{v_1, v_2\}$ is orthogonal. By Thm 4 $\{v_1, v_2\}$ is lin. ind. By Thm 13 §4.5 $\{v_1, v_2\}$ is an orthogonal basis for $\mathbb{C}^2$.

Take $y = \begin{bmatrix} 1+i \\ 0 \end{bmatrix}$.

By Thm 5

$$\begin{bmatrix} 1+i \\ 0 \end{bmatrix} = \frac{\left\langle \begin{bmatrix} 1+i \\ 0 \end{bmatrix}, \begin{bmatrix} i \\ i \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} i \\ i \end{bmatrix}, \begin{bmatrix} i \\ i \end{bmatrix} \right\rangle} \begin{bmatrix} i \\ i \end{bmatrix} + \frac{\left\langle \begin{bmatrix} 1+i \\ 0 \end{bmatrix}, \begin{bmatrix} 2i \\ -2i \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 2i \\ -2i \end{bmatrix}, \begin{bmatrix} 2i \\ -2i \end{bmatrix} \right\rangle} \begin{bmatrix} 2i \\ -2i \end{bmatrix}$$

$$= \frac{(1+i)\overline{(i)}}{1^2 + 1^2} \begin{bmatrix} i \\ i \end{bmatrix} + \frac{(1+i)\overline{(2i)}}{2^2 + 2^2} \begin{bmatrix} 2i \\ -2i \end{bmatrix}$$

$$= \frac{(1+i)(-i)}{2} \begin{bmatrix} i \\ i \end{bmatrix} + \frac{(1+i)(-2i)}{8} \begin{bmatrix} 2i \\ -2i \end{bmatrix}$$

$$= \frac{1-i}{2} \begin{bmatrix} i \\ i \end{bmatrix} + \frac{2-2i}{8} \begin{bmatrix} 2i \\ -2i \end{bmatrix}$$

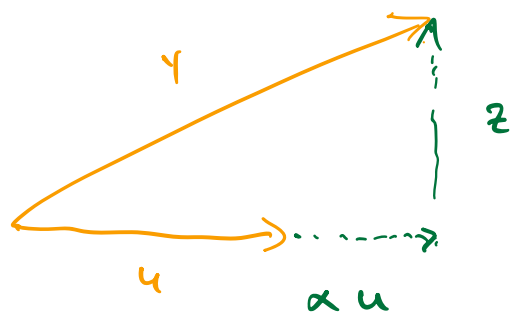
If $B = \{v_1, v_2\}$ then $p_B\left(\begin{bmatrix} 1+i \\ 0 \end{bmatrix}\right) = \begin{bmatrix} \frac{1-i}{2} \\ \frac{2-2i}{8} \end{bmatrix} \in \mathbb{C}^2$.

Orthogonal projections

Suppose $u \in \mathbb{R}^n$ (or $\mathbb{C}^n$) is nonzero and $y \in \mathbb{R}^n$ (or $\mathbb{C}^n$). Want to write

$$y = \alpha u + z$$

where $\alpha \in \mathbb{R}$ (or $\mathbb{C}$) and $\langle z, u \rangle = 0$.



To do this, observe that $z = y - \alpha u$

Since we want $\langle z, u \rangle = 0$, let's compute

$$0 = \langle z, u \rangle = \langle y - \alpha u, u \rangle$$

$$\Rightarrow \quad 0 = \langle y, u \rangle - \alpha \langle u, u \rangle$$

$$\Rightarrow \quad 0 = \langle y, u \rangle - \alpha \|u\|^2$$

$$\Rightarrow \quad \alpha = \frac{\langle y, u \rangle}{\|u\|^2}. \qquad (u \neq 0 \Rightarrow \|u\| \neq 0)$$

We just proved that there is a unique α (above) that gives us $y = \alpha u + z$ with $\langle z, u \rangle = 0$.

Namely $y = \frac{\langle y, u \rangle}{\|u\|^2} u + z.$

Def. Suppose u in $\mathbb{R}^n$ (or $\mathbb{C}^n$) is any nonzero vector and $y \in \mathbb{R}^n$ (or $\mathbb{C}^n$). Then the <u>orthogonal projection</u> of y onto u is

$$\text{proj}_u y = \frac{\langle y, u \rangle}{\langle u, u \rangle} u = \frac{\langle y, u \rangle}{\|u\|^2} u,$$

and $y = \text{proj}_u y + z$ with $\langle z, u \rangle = 0$.